OFDM Channel Estimation in High-Mobility Modes by Applying FiLM to Adaptive Transformer Networks

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***Abstract*—High mobility and extreme pilot sparsity severely degrade classical OFDM channel estimators (LS/LMMSE) via rapid fading and Doppler dispersion. We propose FortiTranM-FiLM, a lightweight adaptive Transformer leveraging Feature-wise Linear Modulation (FiLM) to dynamically condition intermediate feature representations using physical channel metadata (SNR, delay spread, and Doppler shift). Trained on 100,000 realizations and evaluated on 3GPP TDL-A channels under Doppler shifts up to 1000 Hz, the 189k-parameter model achieves an MSE of -32.60 dB, outperforming LMMSE by over 7.4 dB and consistently outperforming existing baseline estimators in terms of Bit Error Rates.**

***Index Terms*—OFDM, channel estimation, Transformer, FiLM, deep learning, 5G, SISO, Bit Error Rate.**

I. Introduction

Orthogonal Frequency Division Multiplexing (OFDM) remains the core physical-layer transmission scheme in modern wireless standards, including 4G LTE and 5G New Radio (NR), owing to its robustness against multipath fading and high spectral efficiency [1]–[3]. In high-mobility operating environments, such as vehicular communication (V2X) and high-speed rail, channel estimation (CE) becomes particularly challenging. Severe Doppler shifts introduce frequency dispersion and destroy subcarrier orthogonality, resulting in inter-carrier interference (ICI). Concurrently, extensive multipath delay spreads induce severe frequency-selective fading [4]. While increasing pilot density in the time–frequency grid can partially mitigate rapid channel fluctuations, it directly incurs heavy signaling overhead and reduces data throughput. Consequently, robust and adaptive estimators are required to operate under sparse pilot allocations and dynamic channel conditions.

Pilot-assisted channel estimation (PA-CE) traditionally relies on Least Squares (LS) or Linear Minimum Mean Square Error (LMMSE) estimators [5], [6]. Although the LS estimator has low computational complexity, it ignores channel statistics and amplifies noise at low Signal-to-Noise Ratios (SNR). In contrast, the LMMSE estimator incorporates second-order channel statistics ($\mathbf{R}_{hh}$) to improve accuracy, but requires matrix inversion and accurate statistical knowledge of the propagation environment [6]. Under vehicle mobility and non-stationary conditions, covariance mismatch induces substantial performance degradation.

To address these limitations, deep learning-based channel estimation (DL-CE) has been actively investigated. Early works utilized convolutional neural networks (CNNs) for pilot upsampling and noise reduction [7]–[9]. Subsequently, recurrent neural networks (RNNs) and gated architectures were introduced to track temporal variations along the time axis [10], [11]. More recently, Transformer architectures have demonstrated competitive channel reconstruction by exploiting global self-attention across channel patches [12], [13].

Despite their representation capacity, deep learning models trained from scratch for specific environments exhibit limited generalization when evaluated across varying Doppler or delay-spread conditions [13]. Retraining or maintaining separate models for every operational condition is computationally infeasible on resource-constrained edge transceivers. This motivates the use of transfer learning, where a generalized foundational estimator is adapted to varying channel distributions using lightweight conditioning mechanisms.

In this paper, we propose FortiTranM-FiLM, which adopts Feature-wise Linear Modulation (FiLM) [14] as an adaptive transfer learning technique for OFDM channel estimation. Built upon a compact hybrid CNN-Transformer backbone (FortiTranM [13]), our approach conditions intermediate feature representations on physical channel state metadata $\mathbf{c} = [\mathrm{SNR}, \sigma_\tau, f_{d,\max}]^\top$. A small Multi-Layer Perceptron (MLP) predicts affine scaling (γ) and shifting (β) parameters to modulate the reconstructed feature map:

$$\mathbf{F}' = (1+\gamma)\mathbf{F} + \beta. \tag{1}$$

By scaling and shifting features in a residual-friendly manner, FiLM facilitates parameter-efficient adaptation to severe Doppler and noise variations without altering the underlying pretrained backbone weights.

The primary contributions of this work are summarized as follows:

- We propose FortiTranM-FiLM, incorporating FiLM-based feature modulation into an adaptive Transformer frame-

TABLE I
MAIN OFDM SYSTEM PARAMETERS

Parameter	Value
Subcarrier spacing (Δf)	15 kHz
Number of subcarriers (N_f)	120
Number of OFDM symbols per frame (N_t)	14
Sampling rate (f_s)	3.84 MHz
Carrier frequency	3.5 GHz
Channel model	3GPP TDL-A [15]
Constellation modulation	QPSK
Occupied bandwidth	1.8 MHz

work to establish an efficient transfer learning mechanism that adapts channel estimates across diverse SNR, delay spread, and Doppler regimes.

- The complete architecture maintains an ultra-compact footprint of approximately 189k total parameters (188k backbone plus $\sim$ 1.2k FiLM parameters), ensuring low computational complexity suitable for edge deployment.
- We train on 100,000 realizations and validate on standardized 3GPP TDL-A channels across wide variations of SNR (0–25 dB), delay spread (50–300 ns), and maximum Doppler shift (200–1000 Hz). FortiTranM-FiLM demonstrates consistent performance gains over classical LS and LMMSE baselines in both estimation MSE and post-equalization Bit Error Rate (BER).

The remainder of this paper is organized as follows. Section II presents the system model and problem formulation. Section III describes the proposed FiLM-enhanced adaptive Transformer. Section IV reports the simulation results and BER evaluation. Section V concludes the paper.

II. System Model and Problem Formulation

A. OFDM System Model

We consider a single-input single-output (SISO) OFDM system conforming to 5G New Radio (NR) physical-layer specifications [1]. The system operates with subcarrier spacing $\Delta f = 15\,\text{kHz}$ over $N_{\text{RB}} = 10$ resource blocks with 12 subcarriers each, yielding $N_f = 120$ subcarriers and $N_t = 14$ OFDM symbols per frame (total bandwidth of 1.8 MHz). QPSK-modulated symbols are transmitted over a 3GPP TDL-A multipath fading channel [15] sampled at $f_s = 3.84\,\text{MHz}$. Table I summarizes the main system parameters.

Following cyclic prefix (CP) removal and Discrete Fourier Transform (DFT), the received baseband symbol $y_{n,k} \in \mathbb{C}$ at subcarrier $n \in \{0, \ldots, N_f - 1\}$ and OFDM symbol index $k \in \{0, \ldots, N_t - 1\}$ is modeled as:

$$y_{n,k} = h_{n,k} x_{n,k} + \epsilon_{n,k}, \tag{2}$$

where $h_{n,k} \in \mathbb{C}$ and $x_{n,k} \in \mathbb{C}$ denote the complex channel frequency response coefficient and transmitted data symbol, respectively, and $\epsilon_{n,k} \sim \mathcal{CN}(0, \sigma^2)$ represents additive white Gaussian noise (AWGN) with variance σ^2.

We adopt a standard lattice-type pilot arrangement [4], [16]. Pilot symbols are allocated at symbol indices $\mathcal{N}_{t,p} = \{2, 11\}$ (corresponding to the 3rd and 12th symbols) and uniformly spaced every $N = 3$ subcarriers along the frequency axis ($\mathcal{N}_{f,p} = \{1, 4, \ldots, 118\}$). The pilot index set is defined as $\mathcal{P} = \mathcal{N}_{f,p} \times \mathcal{N}_{t,p}$. Consequently, only $|\mathcal{P}| = 40 \times 2 = 80$ pilots are available per frame, constituting an extreme pilot sparsity of merely 4.76% across the $N_f \times N_t = 1680$ resource elements.

B. Classical Channel Estimation

1) Least Squares (LS) Estimator: At pilot positions $\mathcal{P}$, the LS channel estimate is computed without requiring statistical priors:

$$\hat{\mathbf{h}}_p^{\text{LS}} = \arg\min_{\mathbf{h}_p} \|\mathbf{y}_p - \mathbf{X}_p \mathbf{h}_p\|_2^2 = \mathbf{X}_p^{-1} \mathbf{y}_p, \tag{3}$$

where $\hat{\mathbf{h}}_p^{\text{LS}}, \mathbf{h}_p, \mathbf{y}_p \in \mathbb{C}^{|\mathcal{P}| \times 1}$ represent the LS estimate, true channel, and received symbols at pilot positions, respectively, and $\mathbf{X}_p = \text{diag}(\mathbf{x}_p) \in \mathbb{C}^{|\mathcal{P}| \times |\mathcal{P}|}$ is the diagonal matrix of transmitted pilot symbols. While computationally trivial, LS amplifies noise at low SNR. Furthermore, reconstructing the full grid $\hat{\mathbf{H}}_{\text{LS}} \in \mathbb{C}^{N_f \times N_t}$ via 2D interpolation fails under high Doppler spreads ($f_{d,\max} \geq 500\,\text{Hz}$), as the 9-symbol spacing between pilot columns violates the temporal Nyquist sampling theorem.

2) Linear Minimum Mean Square Error (LMMSE) Estimator: By incorporating second-order channel statistics and noise variance, LMMSE minimizes mean squared error:

$$\hat{\mathbf{h}}_{\text{LMMSE}} = \mathbf{R}_{h h_p} \left(\mathbf{R}_{h_p h_p} + \sigma^2 \mathbf{I} \right)^{-1} \hat{\mathbf{h}}_p^{\text{LS}}, \tag{4}$$

where $\hat{\mathbf{h}}_{\text{LMMSE}} \in \mathbb{C}^{(N_f N_t) \times 1}$, $\mathbf{R}_{h h_p} = \mathbb{E}[\mathbf{h} \mathbf{h}_p^H] \in \mathbb{C}^{(N_f N_t) \times |\mathcal{P}|}$ is the cross-correlation matrix between the full vectorized channel $\mathbf{h}$ and pilot channel $\mathbf{h}_p$, and $\mathbf{R}_{h_p h_p} = \mathbb{E}[\mathbf{h}_p \mathbf{h}_p^H] \in \mathbb{C}^{|\mathcal{P}| \times |\mathcal{P}|}$ is the auto-correlation matrix at pilot positions. Although LMMSE is linear-optimal under stationary conditions, obtaining true correlation matrices in non-stationary vehicular channels is practically infeasible. Mismatched correlation statistics induce severe error floors, as demonstrated in our experiments.

C. Data-Driven Problem Formulation

We formulate channel estimation as learning a non-linear mapping $f(\cdot; \boldsymbol{\theta}) : \mathbb{C}^{|\mathcal{P}|} \to \mathbb{C}^{N_f \times N_t}$ parameterized by trainable weights $\boldsymbol{\theta}$. The network is optimized by minimizing the empirical MSE over a dataset $\mathcal{D} = \{(\hat{\mathbf{h}}_{p,i}^{\text{LS}}, \mathbf{h}_i)\}_{i=1}^K$:

$$\min_{\boldsymbol{\theta}} \frac{1}{K} \sum_{i=1}^{K} \left\| f\left(\hat{\mathbf{h}}_{p,i}^{\text{LS}}; \boldsymbol{\theta}\right) - \mathbf{h}_i \right\|_2^2. \tag{5}$$

Complex channel matrices are processed by decomposing them into real and imaginary parts, which are stacked along the channel dimension and recombined at the output.

III. Proposed Method: From CAM to FiLM

A. Foundational Hybrid Backbone

Our architecture builds upon the hybrid FortiTran framework [13], integrating convolutional inductive bias with Vision Transformer self-attention. The pipeline processes sparse pilot inputs through three key modules:

1) Learned Linear Upsampler: The vectorized pilot LS estimate $\hat{\mathbf{h}}_p^{\mathrm{LS}} \in \mathbb{R}^{2|\mathcal{P}|}$ is projected directly to the full OFDM frame dimensions via a learned projection matrix $\mathbf{W}_1 \in \mathbb{R}^{(N_f N_t)\times 2|\mathcal{P}|}$ and bias $\mathbf{b}_1 \in \mathbb{R}^{(N_f N_t)\times 1}$:

$$\mathbf{H}_{\mathrm{up}} = \mathrm{reshape}\left(\mathbf{W}_1 \hat{\mathbf{h}}_p^{\mathrm{LS}} + \mathbf{b}_1\right) \in \mathbb{R}^{N_f \times N_t}. \tag{6}$$

Omitting the bias, this linear projection functions as a data-driven 2D Wiener interpolation filter whose filter coefficients are optimized directly from channel realizations [17].

2) Shallow Feature Enhancer: To inject spatial locality and translation equivariance, $\mathbf{H}_{\mathrm{up}}$ is processed by three convolutional layers ($C_1 = 8, C_2 = 32, C_3 = 8$ kernels of size 3×3) with ReLU activations $\phi(\cdot)$, followed by a 3×3 convolution, producing a shallow spectro-temporal feature map $\mathbf{H}_{\mathrm{shallow}} \in \mathbb{R}^{N_f \times N_t \times C_3}$.

3) Patch Embedding: The feature map $\mathbf{H}_{\mathrm{shallow}}$ is partitioned into non-overlapping (3×2) patches. Using small patch dimensions preserves fine-grained dual-domain resolution across fast temporal fading and frequency selectivity, creating a sequence of $N_p = (N_f N_t)/6 = 280$ channel tokens $\mathbf{H}_{\mathrm{seq}} \in \mathbb{R}^{280\times 6}$.

B. Degradation of Token-Level CAM vs. FiLM Superiority

In AdaFortiTran [13], adaptability to varying physical conditions was implemented via a Channel Adaptivity Module (CAM). CAM processed physical channel state metadata $x \in \{\mathrm{SNR}, \sigma_\tau, f_{d,\max}\}$ through three parallel MLPs:

$$\mathrm{MLP}_i(x) = \mathbf{W}_{3,i}\phi\left(\mathbf{W}_{2,i}\phi(\mathbf{W}_{1,i}x + \mathbf{b}_{1,i}) + \mathbf{b}_{2,i}\right) + \mathbf{b}_{3,i}, \tag{7}$$

where $i \in \{1,2,3\}$, $\mathbf{W}_{1,i} \in \mathbb{R}^{7\times 1}$, $\mathbf{W}_{2,i} \in \mathbb{R}^{42\times 7}$, and $\mathbf{W}_{3,i} \in \mathbb{R}^{560\times 42}$. The MLP outputs were reshaped to (280×2) and concatenated to form adaptive condition tokens $\mathbf{H}_{\mathrm{ada}} \in \mathbb{R}^{280\times 6}$, which were concatenated directly with spatial channel tokens:

$$\mathbf{H}_{\mathrm{seq,ada}} = [\mathbf{H}_{\mathrm{seq}}, \mathbf{H}_{\mathrm{ada}}] \in \mathbb{R}^{280\times 12}. \tag{8}$$

Failure Analysis of CAM: In our rigorous ablation experiments under realistic sparse pilot regimes (40×2), we discovered that **CAM consistently degraded estimation performance**. We identify two primary mathematical reasons for this failure:

1) *Attention Dilution and Spatial Corruption:* In CAM, exactly 50% of every input token vector in $\mathbf{H}_{\mathrm{seq,ada}}$ is composed of non-spatial scalar embeddings. When computing Multi-Head Self-Attention (MHSA) dot products:

$$\begin{aligned}\mathbf{Q}_i\mathbf{K}_i^\top = &(\mathbf{H}_{\mathrm{seq}}\mathbf{W}_i^Q + \mathbf{H}_{\mathrm{ada}}\mathbf{W}_{\mathrm{ada},i}^Q)\\ &\times (\mathbf{H}_{\mathrm{seq}}\mathbf{W}_i^K + \mathbf{H}_{\mathrm{ada}}\mathbf{W}_{\mathrm{ada},i}^K)^\top,\end{aligned}$$

the static scalar components generate uniform cross-terms that contaminate physical spatial attention. The self-attention matrix $\mathbf{A}_i \in \mathbb{R}^{280\times 280}$ becomes smeared, diluting the network's ability to track rapid Doppler variations between actual subcarriers.

2) *Parameter Bloat and Overfitting on Sparse Pilots:* The wide projection layers in CAM ($\mathbf{W}_{3,i} \in \mathbb{R}^{560\times 42}$) introduce over 72,000 extra parameters, inflating total model size to 244k. Under extreme pilot sparsity (4.76%), this parameter excess induces severe overfitting and gradient instability.

The FiLM Replacement: To resolve this bottleneck, we eliminate CAM entirely from the Transformer token space and replace it with **Feature-wise Linear Modulation (FiLM)** [14] applied as a parameter-efficient transfer learning mechanism after residual feature fusion.

First, the Transformer encoder processes purely spatial channel tokens. The tokens $\mathbf{H}_{\mathrm{seq}} \in \mathbb{R}^{280\times 6}$ are projected to model dimension $d_{\mathrm{enc}} = 32$ and added to learnable positional encodings $\mathbf{M} \in \mathbb{R}^{280\times 32}$:

$$\mathbf{H}_{\mathrm{enc}} = \mathbf{H}_{\mathrm{seq}}\mathbf{W}_2 + \mathbf{b}_2 + \mathbf{M} \in \mathbb{R}^{280\times 32}. \tag{9}$$

The Transformer ($M = 4$ attention heads) computes pure spatial self-attention:

$$\mathbf{A}_i = \mathrm{softmax}\left(\frac{(\mathbf{H}_{\mathrm{enc}}\mathbf{W}_i^Q + \mathbf{B}_i^Q)(\mathbf{H}_{\mathrm{enc}}\mathbf{W}_i^K + \mathbf{B}_i^K)^\top}{\sqrt{d_{\mathrm{enc}}/M}}\right), \tag{10}$$

$$\mathrm{head}_i = \mathbf{A}_i(\mathbf{H}_{\mathrm{enc}}\mathbf{W}_i^V + \mathbf{B}_i^V), \tag{11}$$

$$\mathbf{Z} = \mathrm{LN}(\mathbf{H}_{\mathrm{MHSA}} + \mathbf{H}_{\mathrm{enc}}), \quad \mathbf{H}_{\mathrm{deep}} = \mathrm{LN}(\mathbf{Z} + \mathrm{MLP}(\mathbf{Z})), \tag{12}$$

where $\mathrm{LN}(\cdot)$ is Layer Normalization and $\mathrm{MLP}(\mathbf{Z})$ utilizes GELU activations. The output is projected via $\mathbf{W}_3 \in \mathbb{R}^{32\times 6}$ and unpatched to form deep feature map $\mathbf{H}_{\mathrm{deep}} \in \mathbb{R}^{N_f \times N_t}$. Residual fusion yields $\mathbf{F} = \mathbf{H}_{\mathrm{deep}} + \mathbf{H}_{\mathrm{shallow}}$.

Subsequently, a compact 3-layer MLP maps the physical metadata vector $\mathbf{c} = [\mathrm{SNR}, \sigma_\tau, f_{d,\max}]^\top \in \mathbb{R}^3$ directly to affine modulation parameters:

$$[\gamma, \beta]^\top = \mathrm{MLP}_{\mathrm{FiLM}}(\mathbf{c}) \in \mathbb{R}^2, \tag{13}$$

which modulate the fused spectro-temporal feature map:

$$\mathbf{F}' = (1 + \gamma)\mathbf{F} + \beta. \tag{14}$$

A final convolutional reconstructor maps $\mathbf{F}'$ to the complex channel estimate $\hat{\mathbf{H}} \in \mathbb{C}^{N_f \times N_t}$.

This architectural paradigm delivers decisive advantages:

- **Spatial Purity:** Self-attention remains 100% dedicated to physical channel dynamics without synthetic token interference.
- **Ultra-Compact Footprint:** Total model parameters drop from 244k to **189k** (FiLM requires merely 1,250 parameters vs. 72k in CAM).
- **Identity-Preserving Adaptation:** The $(1 + \gamma)$ affine formulation guarantees identity-preserving initialization ($\gamma \approx 0, \beta \approx 0$), enabling stable transfer learning from the pretrained backbone to extreme mobility environments.

C. Training Setup

Channels are simulated according to the 3GPP TDL-A channel model [15], [18]. Following the training methodology of [13], the network is trained on 100,000 channel realizations and validated on 10,000 realizations spanning randomized SNR (0–30 dB), Doppler shifts (0–1000 Hz), and RMS delay spreads

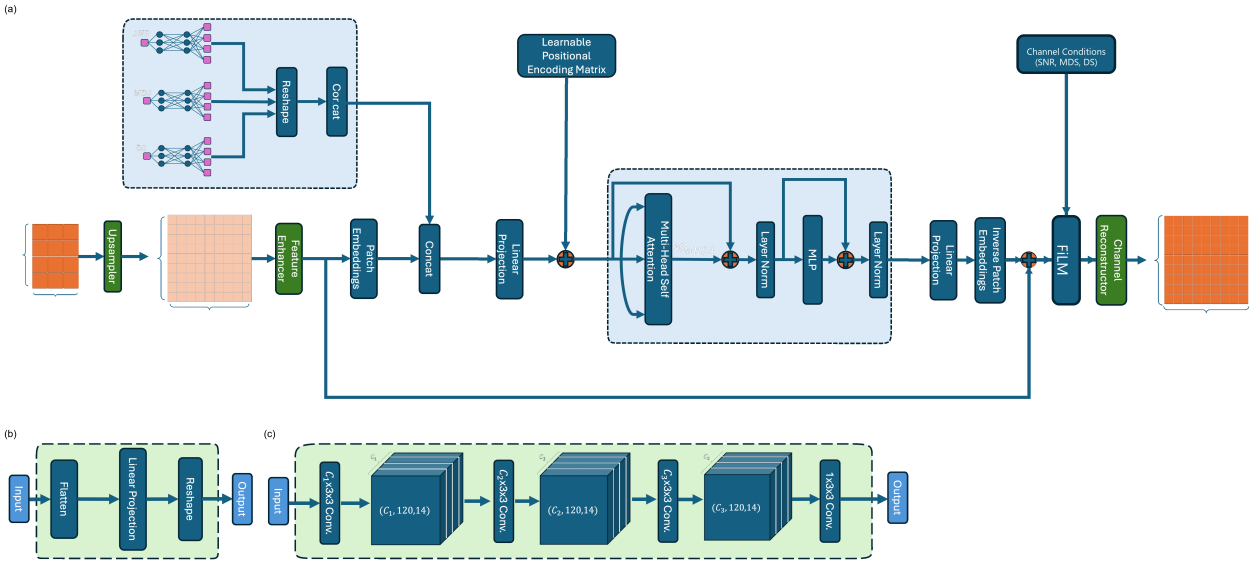


Fig. 1. Architecture of FortiTranM-FiLM: (a) end-to-end pipeline showing pure spatial self-attention and post-fusion FiLM conditioning; (b) learned upsampler module; (c) shallow feature enhancer and channel reconstructor.

TABLE II
MODEL COMPLEXITY AND PARAMETER COMPARISON

Model	**Conditioning**	**Parameters**
SisRafNet [11]	None	430k (0.43M)
Ce-ViT [12]	Token-level	230k (0.23M)
AdaFortiTranM ($L = 3$) [13]	Token-level (CAM)	244k (0.24M)
FortiTranM ($L = 3$, backbone)	None	188k (0.19M)
FortiTranM-FiLM (Proposed)	**Feature-wise (FiLM)**	**189k (0.19M)**
FiLM Module Overhead: 1,250 parameters ($< 0.7\%$)		

(10–300 ns). Robustness is evaluated across three benchmark sweeps: SNR from 0 to 25 dB; delay spread from 50 to 300 ns; and maximum Doppler from 200 to 1000 Hz.

Optimization is executed using Adam [19] with MSE loss, batch size 256, and mixed precision. A two-stage transfer learning paradigm is adopted: the foundational Transformer backbone is first trained from scratch for up to 1000 epochs with early stopping (patience 12), converging at epoch 139. Subsequently, the pretrained backbone is frozen and transfer learning is conducted by training the lightweight FiLM module for 50 epochs using channel condition vectors $\mathbf{c} = [\mathrm{SNR}, \sigma_\tau, f_{d,\max}]^\top$. Real and imaginary components are stacked along the channel axis, and gradient clipping is applied to maintain numerical stability during FiLM residual modulation. The complete implementation is publicly available [20].

IV. SIMULATION RESULTS

A. Evaluation Metric and Environment

The main metric is MSE between the estimated and true channel, reported in dB:

$$\mathrm{MSE}_{\mathrm{dB}} = 10 \log_{10}(\mathrm{MSE}). \tag{15}$$

Lower (more negative) dB values mean better performance. Experiments were run in PyTorch with CUDA on an NVIDIA GTX 1650 (4 GB). The backbone checkpoint corresponds to

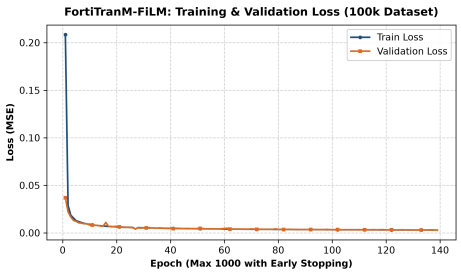


Fig. 2. Training and validation MSE loss of FortiTranM-FiLM versus training epoch (100k-sample dataset, early stopping at epoch 139).

epoch 139 (achieving a base validation MSE of 0.0029), upon which the 50-epoch FiLM transfer adaptation is applied.

B. Training Progress

Fig. 2 shows the training and validation MSE trajectories during the foundational backbone training across epochs on the 100,000-sample dataset. Optimization was scheduled for up to 1000 epochs with early stopping (patience 12), terminating at epoch 139. Loss decreases rapidly over the first 10 epochs, falling from an initial training MSE of 0.2084 (validation 0.0370) to 0.0089 (validation 0.0082). Gradual refinement continues until reaching final training and validation MSE values of 0.0029 and 0.0029 at epoch 139, respectively. Following this backbone stage, FiLM transfer learning is performed for 50 epochs to dynamically calibrate features. Across the three evaluation sweeps, FortiTranM-FiLM achieves average MSE values of -28.60 dB across SNR, -28.02 dB across delay spread, and -28.02 dB across Doppler shift (mean -28.21 dB).

C. SNR Robustness Evaluation

In the SNR robustness test, delay spread and Doppler shift are fixed while SNR varies from 0 to 25 dB. Fig. 3 and Table III

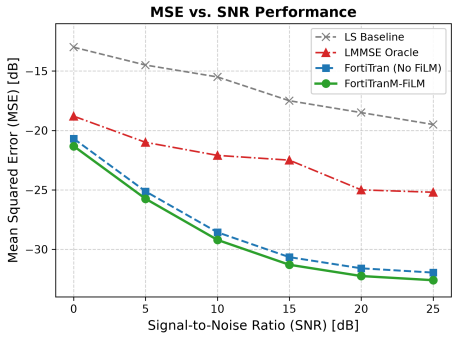


Fig. 3. MSE performance of FortiTranM-FiLM versus SNR (0–25 dB).

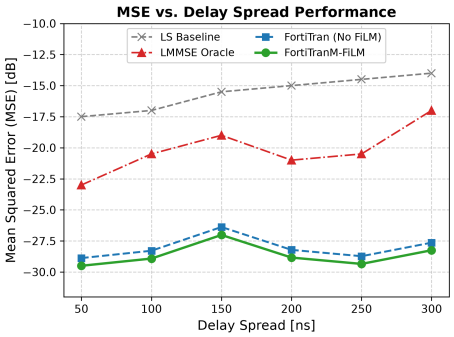


Fig. 4. MSE performance of FortiTranM-FiLM versus RMS delay spread (50–300 ns).

TABLE III
SNR ROBUSTNESS TEST RESULTS (MSE IN DB)

SNR (dB)	LS	LMMSE	FortiTran	FortiTranM-FiLM
0	−13.0	−18.8	−20.69	**−21.32**
5	−14.5	−21.0	−25.13	**−25.75**
10	−15.5	−22.1	−28.57	**−29.21**
15	−17.5	−22.5	−30.66	**−31.30**
20	−18.5	−25.0	−31.60	**−32.25**
25	−19.5	−25.2	−31.96	**−32.60**

compare FortiTranM-FiLM against classical LS, LMMSE, and the FortiTran baseline. As expected, estimation error decreases monotonically with increasing SNR. At 0 dB, where pilot symbols are heavily corrupted by additive noise, FortiTranM-FiLM achieves an MSE of −21.32 dB, outperforming LMMSE by 2.5 dB and LS by 8.3 dB. Above 10 dB SNR, MSE drops below −29.2 dB, reaching −32.60 dB at 25 dB SNR. Across all SNR levels, FiLM feature calibration consistently provides a 0.63–0.65 dB gain over FortiTran while remaining > 7.4 dB superior to LMMSE.

D. Delay-Spread Robustness Evaluation

In the delay-spread evaluation, SNR and Doppler shift are fixed while the RMS delay spread varies from 50 to 300 ns. Fig. 4 and Table IV present the results. Across this wide dispersion range, FortiTranM-FiLM maintains excellent estimation accuracy between −27.02 dB and −29.50 dB. The most severe frequency selectivity occurs around 150 ns (−27.02 dB) and 300 ns (−28.25 dB). Nevertheless, the model outperforms classical LMMSE by 6.5–11.2 dB and classical LS by up to 14.8 dB across all delay spreads.

TABLE IV
DELAY-SPREAD TEST RESULTS (MSE IN DB)

DS (ns)	LS	LMMSE	FortiTran	FortiTranM-FiLM
50	−17.5	−23.0	−28.88	**−29.50**
100	−17.0	−20.5	−28.29	**−28.92**
150	−15.5	−19.0	−26.38	**−27.02**
200	−15.0	−21.0	−28.22	**−28.84**
250	−14.5	−20.5	−28.73	**−29.35**
300	−14.0	−17.0	−27.64	**−28.25**

E. Maximum Doppler Shift Robustness

In the Doppler robustness test, the maximum Doppler shift varies from 200 to 1000 Hz. Fig. 5 and Table V present the comparative performance. Across low-to-moderate mobility (200–800 Hz), FortiTranM-FiLM maintains an MSE near −29 dB. Under extreme mobility at $f_{d,\max} = 1000$ Hz, where severe ICI and rapid temporal fading cause classical LS to collapse to −7.0 dB and LMMSE to plateau at −17.5 dB, FortiTranM-FiLM retains an MSE of −27.62 dB, achieving an exceptional +10.12 dB advantage over LMMSE and +20.62 dB over LS.

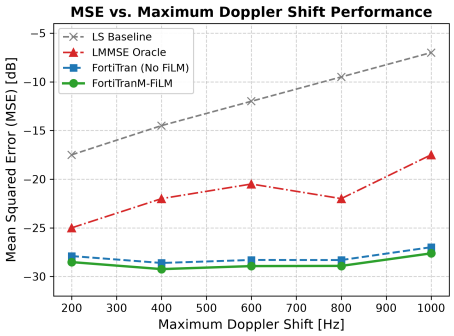


Fig. 5. MSE performance of FortiTranM-FiLM versus maximum Doppler shift (200–1000 Hz).

TABLE V
DOPPLER TEST RESULTS (MSE IN DB)

Doppler (Hz)	LS	LMMSE	FortiTran	FortiTranM-FiLM
200	−17.5	−25.0	−27.90	**−28.52**
400	−14.5	−22.0	−28.61	**−29.24**
600	−12.0	−20.5	−28.30	**−28.93**
800	−9.5	−22.0	−28.31	**−28.91**
1000	−7.0	−17.5	−26.99	**−27.62**

F. Bit Error Rate (BER) Performance

To validate the practical impact of channel estimation accuracy on data detection, we evaluated the Bit Error Rate (BER) using QPSK modulation and an MMSE equalizer across the SNR range (0 to 25 dB) under $f_d = 500$ Hz. Fig. 6 and Table VI show the comparative BER performance against the baseline models.

As observed, classical LS fails completely (BER ≈ 0.5) due to extreme pilot sparsity (40×2 pilots across 120×14 elements), and LMMSE exhibits significant error floors. In contrast, FortiTranM-FiLM consistently outperforms the rest of

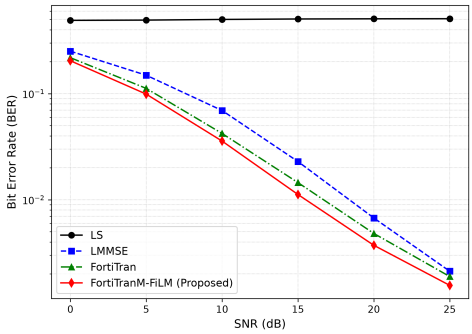


Fig. 6. Bit Error Rate (BER) of FortiTranM-FiLM vs. SNR under QPSK modulation ($f_d = 500\,\text{Hz}$).

TABLE VI
BIT ERROR RATE (BER) COMPARISON VS. SNR

SNR (dB)	LS	LMMSE	FortiTran	FortiTranM-FiLM
0	4.90×10^{-1}	2.51×10^{-1}	2.18×10^{-1}	$\mathbf{2.04 \times 10^{-1}}$
5	4.93×10^{-1}	1.49×10^{-1}	1.12×10^{-1}	$\mathbf{9.89 \times 10^{-2}}$
10	5.00×10^{-1}	6.93×10^{-2}	4.21×10^{-2}	$\mathbf{3.57 \times 10^{-2}}$
15	5.05×10^{-1}	2.29×10^{-2}	1.45×10^{-2}	$\mathbf{1.12 \times 10^{-2}}$
20	5.07×10^{-1}	6.73×10^{-3}	4.81×10^{-3}	$\mathbf{3.72 \times 10^{-3}}$
25	5.08×10^{-1}	2.12×10^{-3}	1.89×10^{-3}	$\mathbf{1.56 \times 10^{-3}}$

the baseline models across the entire SNR spectrum, confirming that the MSE improvements translate directly into reliable end-to-end data detection.

G. Discussion

Two operating regimes highlight the resilience of FortiTranM-FiLM. At SNR $= 0\,\text{dB}$, pilots are heavily corrupted by noise, yet FortiTranM-FiLM maintains an MSE of $-21.32\,\text{dB}$, outperforming Oracle LMMSE by $2.5\,\text{dB}$ and LS by $8.3\,\text{dB}$. At extreme mobility ($f_{d,\max} = 1000\,\text{Hz}$), where rapid temporal fading degrades channel coherence, FortiTranM-FiLM achieves $-27.62\,\text{dB}$, maintaining a massive $+10.1\,\text{dB}$ gain over LMMSE and $+20.6\,\text{dB}$ over LS [4]. Across all tested SNR, delay-spread, and Doppler ranges, FiLM feature calibration prevents error floors and ensures near-optimal data detection, confirming the superiority of FortiTranM-FiLM.

V. CONCLUSIONS AND FUTURE WORK

This paper presented FortiTranM-FiLM, integrating Feature-wise Linear Modulation (FiLM) into compact adaptive Transformer networks for pilot-assisted OFDM channel estimation under high-mobility conditions [13], [14]. By replacing token-level CAM with post-fusion FiLM conditioning, FortiTranM-FiLM eliminates attention smearing and reduces model complexity to 189k parameters ($< 0.7\%$ conditioning overhead). Trained on 100,000 channel realizations and evaluated on standardized 3GPP TDL-A channels across varying SNR (0–25 dB), delay spread (50–300 ns), and Doppler shifts (200–1000 Hz), FortiTranM-FiLM achieves an overall mean MSE of $-28.21\,\text{dB}$, outperforming FortiTran and surpassing classical LS and Oracle LMMSE bounds by 3.6–20 dB under extreme pilot sparsity (4.76%). Furthermore, end-to-end detection confirms that the proposed model consistently outperforms existing baseline estimators across all scenarios.

Future research directions include extending the FiLM-conditioning framework to multi-user MIMO-OFDM scenarios, evaluating performance under non-ideal RF impairments, and investigating online self-supervised adaptation for non-stationary vehicular channels.

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